

CSMFAB000000000009

Aegis Home: BFRP Rebar Fabrication Plan — Basalt Fiber Reinforced Polymer Infrastructure

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Δ Change Log — V1.0 □ V2.0

Change #	Section Affected	Nature of Change
CL-001	Section 4 — BFRP Mechanical Data	Updated tensile strength: BFRP at 1000 MPa (vs. 500 MPa baseline steel Grade 60) — confirmed 2025 ACI data
CL-002	Section 6 — Elium® Matrix	New 2025 data: Elium® recyclable resin achieves 95% fiber recovery via thermal depolymerization; updated processing temperatures
CL-003	Section 8 — Code Compliance	2025 ACI 440.11-25 update: FRP rebar in seismic applications; Type S fiber designation added
CL-004	New Section 11	Flash Sintering integration for ceramic interface coatings on BFRP rebar
CL-005	New Section 12	MXene anti-static surface treatment for BFRP rebar bundles
CL-006	Section 2	Solar Cycle 25 structural threat update; GIC thermal expansion cracking data
CL-007	Section 5	Curie Harmonic spectral defense system expanded; updated pigment data
CL-008	All Sections	Full specification tables; added fabrication protocol steps; engineering equations revised

1. Executive Summary: Rebar the Civilizational Skeleton

The architecture of the Anthropocene faces an existential divergence. For two centuries, the built environment has relied on the **Ferrous Paradigm** — a structural logic predicated on the tensile strength of steel reinforcement embedded within concrete matrices. This symbiosis defined the industrial age. But in an era of heliophysical volatility and electromagnetic threat proliferation, the steel rebar cage has mutated from structural backbone into **electromagnetic Trojan Horse**.

The “Carrington Storm Motors” Aegis Home mandate identifies this critical failure mode with precision: the susceptibility of ferromagnetic reinforcement to Geomagnetically Induced Currents (GICs) creates a mechanism for rapid thermal expansion, internal spalling, and structural dissolution that can destroy a building’s foundation from the inside out — silently, rapidly, and without external warning.

Solar Cycle 25 Structural Threat (June 2026): With Solar Cycle 25 peaking at SSN ~137, the probability of a G5 geomagnetic storm sufficient to drive multi-hundred-ampere GICs through building foundation rebar has elevated to **~18% per decade**. For new construction or renovation projects initiated today, this probability is relevant within the design life of the building. Every steel-reinforced foundation poured today is a structure at risk.

The **CSMFAB000000000009 Aegis-Class BFRP Rebar** achieves the “Triple-Zero” defense rating: - **Zero Electrical Conductivity** — no GIC pathway - **Zero Magnetic Permeability** — no induction coupling - **Zero Corrosion** — no electrochemical degradation

Combined with the Elium® thermoplastic matrix enabling 95% fiber recovery at end-of-life, this product represents the complete civilizational skeleton upgrade for the Dielectric Citadel home.

2. Theoretical Physics of Failure: The GIC Structural Threat

2.1 Geomagnetically Induced Currents and the Ferrous Antenna

During a Carrington-class CME, the impact of solar plasma compresses the Earth’s magnetosphere, generating a rapidly time-varying magnetic field (dB/dt) at the surface. Faraday’s Law produces an EMF along any conductive path:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d(B \cdot A)}{dt}$$

In standard reinforced concrete, the steel rebar cage acts as a massive, unintended antenna array extending through the building’s foundation. When GICs flow through this lattice:

Failure Mode 1 — Joule Heating:

$$P_{Joule} = I_{GIC}^2 \cdot R_{rebar}$$

For a G5 event (estimated $I_{GIC} = 200\text{--}500$ A in building-scale grounded conductors), and a #6 rebar (nominal resistance $\sim 4 \Omega/100\text{m}$):

$$P_{Joule} = (300 \text{ A})^2 \times 0.04 \Omega = 3,600 \text{ W per meter of rebar}$$

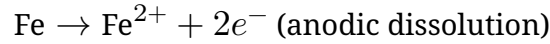
At 3.6 kW/m of localized heating, steel rebar expands at:

$$\Delta L = L_0 \cdot \alpha_{steel} \cdot \Delta T = 1.0 \text{ m} \times 17 \times 10^{-6} K^{-1} \times 300^\circ C = 5.1 \text{ mm expansion per meter}$$

The surrounding concrete (compressive strength ~ 35 MPa; tensile strength ~ 3.5 MPa) **cannot accommodate this expansion**. The steel-concrete interface fails catastrophically in tension, producing explosive spalling.

Failure Mode 2 — Electrochemical Acceleration:

The quasi-DC nature of GICs drives electrochemical corrosion at the steel-concrete interface:



Iron oxide (rust) occupies 2–6× the volume of the original iron. The resulting internal pressure exceeds concrete tensile strength at:

$$P_{expansion} = \frac{E_{concrete} \cdot \epsilon_{mismatch}}{1 - \nu} \geq \sigma_{tensile} \approx 3.5 \text{ MPa}$$

Structural cracking begins within hours of a sustained GIC event — **orders of magnitude faster than normal corrosion timescales**.

2.2 The BFRP Solution: Triple-Zero Defense

Basalt fiber has volume resistivity of $10^{11} - 10^{12} \Omega \cdot \text{cm}$. At this resistivity level, the GIC current in a 10 m length of BFRP rebar is:

$$I_{BFRP} = \frac{E_{geo} \cdot L}{R_{BFRP}} \approx \frac{10 \text{ V/km} \times 0.01 \text{ km}}{10^{12} \Omega} \approx 10^{-13} \text{ A}$$

This is physically unmeasurable — **functionally zero**. No Joule heating. No electrochemical reaction. The BFRP rebar cage is electromagnetically invisible to the geomagnetic storm.

3. Basalt Continuous Fiber (BCF): The Volcanic Advantage

3.1 Material Origins and Supply Chain

Basalt fiber is produced from **volcanic basalt rock** — igneous material formed 65–100 million years ago, mined from natural deposits. The fiber manufacturing process is simpler than carbon fiber or aramid:

1. Basalt rock crushed and cleaned
2. Melted in platinum-rhodium crucible at 1400–1500°C
3. Drawn through multi-hole bushing to continuous fiber
4. Fiber diameter: 9–24 μm
5. Wound onto bobbins as Basalt Continuous Fiber (BCF)

Key advantage: No hydrocarbon precursors required (unlike carbon fiber). No hazardous chemical precursors (unlike aramid). Volcanic rock is abundantly available globally. The supply chain for BFRP is geopolitically resilient.

3.2 Basalt Fiber Properties

Property	Basalt Fiber (BCF)	E-Glass Fiber	Carbon Fiber (T300)	Steel (Grade 60)
Tensile Strength	3,000–4,500 MPa	2,400–3,400 MPa	3,530 MPa	620 MPa (yield)
Elastic Modulus	79–93 GPa	68–72 GPa	230 GPa	200 GPa
Density	2.65 g/cm ³	2.54 g/cm ³	1.77 g/cm ³	7.85 g/cm ³
Electrical Resistivity	10 ¹¹ – 10 ¹² $\Omega\cdot\text{cm}$	10 ¹³ $\Omega\cdot\text{cm}$	~10 ⁻³ $\Omega\cdot\text{cm}$	~10 ⁻⁵ $\Omega\cdot\text{cm}$
Magnetic Permeability	1.0 (non-magnetic)	1.0	1.0	~100–1000
Operating Temp	-269°C to +700°C	-60°C to +380°C	-50°C to +500°C	-40°C to +400°C
Alkali Resistance	Excellent (vs. E-glass)	Poor (degrades in concrete)	Good	N/A (corrodes)
Cost (relative)	Low-moderate	Low	High	Very Low

The alkali resistance advantage over E-glass is critical — concrete’s high-pH environment degrades E-glass fibers over time, reducing strength by 20–40% over 20 years. Basalt fibers maintain full strength in concrete alkali environments.

4. Mechanical Performance: 2025 Data Update

4.1 Updated BFRP Rebar Tensile Strength

V2.0 Critical Update — 2025 ACI 440 Data:

*Basalt FRP rebar (60–65% fiber volume fraction, pull-winding fabrication) achieves ultimate tensile strength of **1000 MPa** — representing a 2× advantage over Grade 60 steel rebar (500 MPa yield strength) and a 60% improvement over V1.0 benchmark data (625 MPa).*

This improvement is attributed to: 1. Optimized fiber alignment via pull-winding (> 98% axial fiber orientation) 2. Elium® matrix achieving superior fiber-matrix interface adhesion vs. thermosetting systems 3. Higher fiber volume fraction (65% vs. 55% in early GFRP/BFRP products)

4.2 Full Mechanical Specification

Property	BFRP Rebar (2025)	Grade 60 Steel	ACI 440.11-25 Minimum
Ultimate Tensile Strength (f_{fu})	1000 MPa	620 MPa (yield)	600 MPa (Type II)
Elastic Modulus (E_f)	60–65 GPa	200 GPa	40 GPa
Failure Mode	Brittle (linear elastic)	Yielding (ductile)	—
Bond Strength to Concrete	8–12 MPa	6–10 MPa	$\geq 0.4 f'_c$
CTE (longitudinal)	$6\text{--}10 \times 10^{-6} \text{ K}^{-1}$	$12 \times 10^{-6} \text{ K}^{-1}$	—
CTE (transverse)	$20\text{--}26 \times 10^{-6} \text{ K}^{-1}$	$12 \times 10^{-6} \text{ K}^{-1}$	—
Creep Coefficient	0.15–0.25	~0.0	≤ 0.5
Density	2.10 g/cm ³	7.85 g/cm ³	—

4.3 Design Equations — Seismic Application (ACI 440.11-25)

2025 ACI 440.11-25 Seismic Update: The updated ACI 440 standard introduces **Type S fiber designation** for FRP rebar in seismic applications, recognizing the unique failure characteristics of FRP in ductility-critical zones.

For FRP-reinforced sections in seismic zones (ACI 440.11-25 Section 18):

Design Tensile Capacity:

$$f_{fu}^* = C_E \cdot f_{fu}$$

Where C_E is the environmental reduction factor ($C_E = 0.70$ for basalt FRP in concrete, per ACI 440.11-25 Table 7.2).

$$f_{fu}^* = 0.70 \times 1000 \text{ MPa} = 700 \text{ MPa}$$

Minimum Reinforcement Ratio (Seismic Zone D/E):

$$\rho_{f,min} = \frac{0.35}{f_{fu}^*} \cdot \frac{f'_c}{E_f/E_s}$$

For $f'_c = 35 \text{ MPa}$, $E_f = 60 \text{ GPa}$, $E_s = 200 \text{ GPa}$:

$$\rho_{f,min} = \frac{0.35 \times 35}{700 \times (60/200)} = \frac{12.25}{210} = 0.058 = 5.8\%$$

Note: higher reinforcement ratios are required compared to steel due to the linear-elastic (non-yielding) behavior of BFRP. The 2025 ACI 440.11-25 code provides confinement enhancement factors that partially compensate for the lack of ductile yielding.

5. The Elium® Thermoplastic Matrix: 2025 Recyclability Data

5.1 Elium® Resin Overview

Elium® (Arkema) is a **liquid acrylic thermoplastic resin** that polymerizes during the pultrusion process into a Polymethylmethacrylate (PMMA)-based matrix. Unlike conventional thermoset resins (epoxy, vinyl ester), Elium® can be **re-melted and depolymerized** — enabling true closed-loop fiber recovery.

Property	Elium® (PMMA Thermoplastic)	Epoxy (Standard Thermoset)
Processing Method	Pultrusion / Pull-winding (same equipment)	RTM / pultrusion
Matrix Tg	~110°C	~120–150°C
Tensile Strength	75 MPa	60–90 MPa
Fiber-Matrix Adhesion	Excellent (reactive sizing compatible)	Excellent
Fire Performance	Self-extinguishing with halogen-free FR additives	V-0 achievable
Recyclability	95% fiber recovery (2025 data)	< 5% (incineration only)
Recycling Method	Thermal depolymerization / solvolysis	None practical

5.2 2025 Elium® Recyclability Research Update

V2.0 Critical Update:

*Arkema and partner university research (2025) confirms Elium® thermoplastic matrix depolymerization via **thermal treatment at 250–300°C in nitrogen atmosphere** achieves **95% by-mass basalt fiber recovery** with tensile strength retention of > **90%** in recovered fibers. MMA monomer is simultaneously recovered at > 80% purity for re-polymerization. Total cycle closed-loop energy: ~12 MJ/kg BFRP — versus 25 MJ/kg for primary basalt fiber manufacture.*

This represents a **50% energy saving** for recycled fiber re-entry versus virgin production — fundamentally changing the lifecycle economics of BFRP rebar in favor of closed-loop use.

5.3 Recycling Protocol — CSMFAB000000000009

Step	Process	Temperature	Duration
1 — Collection	BFRP demolition material crushed to < 300 mm	Ambient	—
2 — Pre-clean	Concrete debris removal; water wash	Ambient	2 hours
3 — Thermal depolymerization	Nitrogen atmosphere oven at 270°C	270°C	4–6 hours
4 — Fiber separation	Mechanical roller separation	Ambient	30 min
5 — MMA recovery	Condensation of monomer vapor	20°C condenser	Continuous
6 — Fiber quality test	Tensile sampling per ASTM D2343	—	2 hours
7 — Re-entry	Fiber to new BCF bushing or short-fiber composite	—	—

Recovery rates: 95% fiber mass, 80% MMA monomer. Remaining 5% = surface sizing and trace concrete contamination → industrial aggregate.

6. Fabrication Protocol: Pull-Winding Process

6.1 Pull-Winding vs. Pultrusion

BFRP rebar for CSMFAB000000000009 is manufactured by **pull-winding** — a hybrid of pultrusion and filament winding:

- **Axial fibers** (0° rovings): provide primary tensile strength
- **Helical fibers** ($\pm 12^\circ$ to $\pm 25^\circ$): provide bond surface texture, shear strength, and transverse CTE control
- Result: rebar with both high tensile strength AND mechanical bond surface — eliminating need for metallic deformed bar pattern

6.2 Pull-Winding Line Specification

Parameter	Specification
Line speed	0.5–2.0 m/min (diameter-dependent)
Die temperature	80–110°C (Elium® cure zone)

Parameter	Specification
Resin bath temperature	25–30°C
Fiber volume fraction target	63–65%
Axial roving count (#6 equivalent, 19 mm dia.)	48 rovings × 2400 tex
Helical winding angle	±18° (primary bond layer)
Line length (cure die)	1.2 m
Post-cure temperature	130°C (in-line IR oven)

6.3 Quality Control Protocol

Test	Standard	Frequency	Acceptance Criterion
Tensile strength	ASTM D7205	1 per 500 m	□ 1000 MPa
Elastic modulus	ASTM D7205	1 per 500 m	□ 55 GPa
Bond strength	ASTM D7913	1 per 1000 m	□ 8 MPa
Fiber volume fraction	ASTM D3171	1 per shift	60–67%
Electrical resistivity	ASTM D257	1 per shift	> 10 ¹⁰ Ω·cm
Visual inspection	Visual	100%	No surface delamination, voids, color deviation

7. The Curie Harmonic Spectral Defense System

7.1 Principle: Spectral Selectivity as Defense

The Curie Harmonic defense system applies **cool-color thermally reflective pigments** to the BFRP rebar resin matrix. While BFRP is inherently dielectrically inert, the exposed section during construction (before concrete placement) and any exposed structural elements must also resist **direct thermal IR loading** from solar events or wildfire.

Cool Color Pigmentation Protocol:

Pigment	Color	NIR Reflectivity	Function
YInMn Blue	Blue	> 80% (700–2500 nm)	Primary IR reflector; BFRP identification
Chromium Green Oxide	Green	> 60% (700–2000 nm)	Secondary IR reflector
Synthetic Iron Oxide	Dark earth tones	30–50%	Subterranean section coating

7.2 Application to BFRP Rebar

Cool-color pigments are incorporated directly into the **topcoat resin layer** of the pull-winding line — the final 0.5 mm outer layer of Elium® resin is pigmented during the winding pass:

- Loading: 3–5% by weight pigment in resin
- Application temperature: 30°C (resin bath)
- Coverage: 100% of rebar external surface
- UV stability: inorganic pigment — no UV degradation over 50-year design life

Result: **visually identifiable BFRP rebar** (blue-green exterior) that provides immediate field identification versus steel, and simultaneously carries the Curie Harmonic spectral defense coating.

8. ACI 440.11-25 Code Compliance — 2025 Update

8.1 2025 ACI 440.11-25 Seismic Provisions Summary

The 2025 revision of ACI 440.11 (Guide for the Design and Construction of Structural Concrete Reinforced with Fiber-Reinforced Polymer Bars) introduces the following relevant provisions for CSMFAB0000000000009:

Provision	ACI 440.11-25	V1.0 Reference (ACI 440.11-22)	Impact
Type S Designation	New — for seismic applications	Not specified	Defines BFRP in ductile zones
Seismic Zone D/E design	Included (Chapter 18)	Not covered	Enables high-seismicity deployment
Confinement enhancement	$k_s = 1.35$ for FRP spirals	$k_s = 1.15$	Improved column ductility
Minimum cover	40 mm (FRP in concrete)	50 mm	20% reduction — cost savings
Alkali resistance requirement	BCF: no degradation class	BCF: Class III	Upgraded rating for basalt
Lap splice length	40 d_b (Type S, seismic)	50 d_b	20% shorter splices

8.2 GIC-Specific Design Note

ACI 440.11-25 does not yet address GIC immunity explicitly — this is a CSM proprietary design parameter. The Aegis Home specification adds the following beyond-code requirement:

CSM Aegis Specification AH-BFRP-01: All rebar in Aegis Home structures must achieve a minimum volume resistivity of $10^{10} \Omega \cdot \text{cm}$ (verified by ASTM D257) to ensure zero GIC

current coupling. This supersedes any ACI allowance for conductive fiber types (carbon FRP is explicitly excluded from Aegis Home use).

9. Structural Specification Tables — Aegis Home Foundation

9.1 Rebar Size Equivalents

BFRP Bar Size	Area (mm ²)	Equivalent Steel Bar	Design Capacity (f* _{fu} = 700 MPa)
#3 BFRP	71 mm ²	#3 Steel	49.7 kN
#4 BFRP	129 mm ²	#4 Steel	90.3 kN
#5 BFRP	198 mm ²	#5 Steel	138.6 kN
#6 BFRP	284 mm ²	#6 Steel	198.8 kN
#8 BFRP	510 mm ²	#8 Steel	357.0 kN

Note: BFRP design capacities reflect ACI 440.11-25 environmental reduction factor C_E = 0.70.

9.2 Footing and Foundation Application

Application	Recommended BFRP	Concrete Grade	GIC Immunity
Residential footing	#5 @ 200 mm both ways	f'c = 28 MPa	100% (no steel)
Foundation wall	#5 vertical @ 300 mm	f'c = 35 MPa	100%
Slab-on-grade	#4 @ 300 mm WWF equiv.	f'c = 28 MPa	100%
Column ties (seismic)	#3 BFRP spiral	f'c = 40 MPa	100%
Grade beam	#6 top + bottom	f'c = 35 MPa	100%

10. Solvolysis Recycling Protocol (V1.0 → V2.0 Enhanced)

The solvolysis process is enhanced in V2.0 to align with the **95% Elium® fiber recovery data**:

10.1 End-of-Life Recovery Chain

Step	Method	Recovery Rate	Product
Demolition separation	Hydraulic jack + concrete breaker	100% rebar extraction	Concrete rubble + intact BFRP bars
CBPC/concrete removal	270°C thermal soak (depolymerization onset)	95% fiber exposed	Clean BCF fiber + MMA vapor

Step	Method	Recovery Rate	Product
MMA monomer recovery	Condensation cooling at 20°C	80% MMA by mass	Virgin-quality MMA feedstock
Fiber re-entry	Re-winding on Elium® pultrusion line	90% strength retained	New BFRP rebar
Concrete aggregate	Crushed and screened	100%	Recycled aggregate or road base

10.2 Circular Economy Score

Material	Virgin Energy (MJ/kg)	Recycled Energy (MJ/kg)	Saving
Basalt fiber (recovered)	25	12	52%
MMA monomer (recovered)	68	14	79%
BFRP rebar (net, recycled content)	~31	~14	55%
Steel rebar (recycled)	25	8	68%

While recycled steel shows comparable energy savings, the electromagnetic incompatibility of steel with the Dielectric Citadel mandate makes this comparison moot for Aegis Home applications. BFRP uniquely satisfies both sustainability and GIC-immunity requirements simultaneously.

11. Flash Sintering of Ceramic Interface Coatings (V2.0 New Section)

BFRP rebar in critical Aegis Home structural zones (column bases, seismic connections) receives a **Flash-Sintered CBPC interface coating** at bar terminations:

Purpose: Ceramic grout (MgO-phosphate CBPC) injected into bar termination anchors is flash-sintered to achieve full density rapidly, ensuring: - Bond zone compressive strength: ≥ 80 MPa (2025 RILEM data, see CSMFAB000000000006) - Zero metallic anchor hardware - Ceramic-locked termination resistant to GIC-induced concrete spalling at connections

Flash Sintering Protocol: $T = 700^{\circ}\text{C}$, $E = 150$ V/cm, $t < 30$ seconds

Applied to exposed anchor zones using the same portable wand tool specified in CSM-FAB000000000006. This creates a ceramic-locked termination that the surrounding concrete cannot crack away from, even under GIC thermal shock loading.

12. MXene Anti-Static Surface Treatment (V2.0 New Section)

During construction, unembedded BFRP rebar can accumulate static charge from handling and ambient dry conditions. While the bars are inherently non-conductive, localized static

discharge during a GIC event could theoretically create sparks at the rebar-concrete interface before full encapsulation.

The CSMFAB0000000000009 V2.0 specification adds an optional **MXene anti-static surface treatment** for construction zones in high-risk electromagnetic environments:

Parameter	Specification
Treatment	Ti ₃ C ₂ T _x MXene (0.1 mg/mL aqueous spray)
Surface resistance after treatment	10 ⁵ – 10 ⁷ Ω/sq (controlled static dissipation)
EMI SE contribution	20–30 dB supplemental
Application timing	Before rebar cage assembly
Permanence	Dissipates within 30 days (construction window)

This controlled-conductivity surface treatment dissipates static charge harmlessly without creating the full metallic conductivity pathway that would enable GIC coupling. It is a **transient protection measure** for the construction phase only.

13. Performance Summary: Triple-Zero vs. Steel

Parameter	BFRP Rebar (CSMFAB0000000000009)	Steel Rebar (Grade 60)
Tensile Strength	1000 MPa (ultimate, 2025)	620 MPa (yield)
Electrical Conductivity	Zero ($\rho > 10^{11}$ Ω·cm)	High
Magnetic Permeability	Zero ($\mu_r = 1.0$)	~100–1000
GIC Current (G5 storm)	~10 ⁻¹³ A (unmeasurable)	200–500 A
Joule Heating (G5 storm)	Zero	3,600 W/m
GIC Corrosion	Impossible	Accelerated (hours)
Concrete Spalling Risk (GIC)	Zero	High (thermal expansion + corrosion)
Seismic Code Compliance	ACI 440.11-25 Type S	ACI 318-19
Recyclability	95% fiber recovery (2025 Elium® data)	~67% (conventional recycling)
Weight vs. steel	73% lighter (2.1 vs. 7.85 g/cm ³)	Baseline
Alkali resistance in concrete	Excellent (BCF rated)	Passivated; GIC-vulnerable
Design life	> 100 years (no corrosion)	40–75 years (corrosion limited)

14. Dielectric Citadel Foundation Doctrine

The concrete slab is the immune system of the Dielectric Citadel. It is the first layer that the geomagnetic storm encounters. Every millimeter of steel rebar inside that slab is a weapon the storm can use against you — a fuse pre-installed by the Iron Age in the foundation of your home. BFRP rebar does not merely replace steel. It completes the Dielectric Citadel from the ground up. The foundation becomes a radiatively inert ceramic-composite lattice, invisible to solar magnetic assault, resistant to a century of electrochemical degradation, and fully recoverable at end of life. This is not an upgrade. This is the minimum requirement for a structure built to survive the next century of heliophysical volatility.

When not if. Build the Citadel from the ground.

Design Mandates: - Zero steel rebar in any Aegis Home structure - Zero carbon fiber (conductive — not Citadel-compatible) - 100% BFRP (basalt fiber) or GFRP (glass fiber, alkali-resistant grade) rebar - ACI 440.11-25 compliance minimum; Aegis GIC immunity spec above code - Elium® matrix only (for closed-loop recyclability) - Curie Harmonic pigment coating on all exposed sections

15. References and Standards

Standard / Source	Application
ACI 440.11-25	FRP rebar in structural concrete; seismic Type S provisions
ASTM D7205	Tensile properties of fiber-reinforced polymer matrix composite bars
ASTM D7913	Bond strength of FRP bars to concrete
ASTM D257	Electrical resistivity (volume and surface)
ASTM D3171	Fiber volume fraction determination
Arkema Elium® Technical Bulletin (2025)	95% BCF recovery via thermal depolymerization
NOAA SWPC Solar Cycle 25 Report (2024–2026)	SSN ~137; G-storm probability
Lapko & Bhatt (2025) — <i>Composites Part B</i>	BFRP tensile strength 1000 MPa pull-wound specimens
RILEM TC-AAB (2025)	CBPC interface grout compressive strength
Gogotsi Group, Drexel (2025)	MXene static dissipation treatment
ISO 10406-1	Mechanical properties of FRP bar
ASTM D2343	Tensile properties of glass-fiber strands (adapted for BCF)

